A Production-Oriented Machine Learning Pipeline for Real Estate Mass Appraisal with Official Geospatial Feature Engineering

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Abstract. *Real estate mass appraisal underpins fair property taxation in Brazilian municipalities, but attributes that curated-dataset models rely on are unavailable in official cadastral systems at inference time. We present a production-oriented pipeline that restricts features to attributes derivable from the municipal IPTU cadastre, standardizes territory through a spatial join with official shapefiles, and uses a large language model offline to encode neighborhoods into ten locational tiers, persisted as a versioned artifact never invoked at serving time. Among three tree-based ensembles evaluated on about* $35{,}000$ *deduplicated listings under stratified* 5*-fold cross-validation with a* $90/10$ *split, CatBoost reaches a holdout mean absolute error near BRL* $222{,}000$ *to* $225{,}000$*,* R^2 *around* 0.89*, and a median absolute percentage error close to* 11%*; coordinates with the language-model group match the full baseline using fewer features.*

1. Introduction

Accurate property valuation is fundamental to the fiscal health of Brazilian municipalities. The *Imposto Predial e Territorial Urbano* (IPTU), the Brazilian municipal property tax, depends on reliable estimates of property market values, yet many cities still rely on outdated generic value tables and manual assessments that fail to capture the spatial and temporal dynamics of the real estate market [Oliveira et al. 2022], resulting in revenue losses, inequitable tax burdens, and reduced trust in municipal governance.

Machine learning methods, particularly gradient boosting, achieve strong performance on heterogeneous tabular data and are widely used for property valuation [Chen and Guestrin 2016]. The Brazilian context, however, poses specific challenges. Real estate data available for operational modeling are fragmented across municipal cadastres and online listing platforms, often with noise, inconsistent geographic references, and missing attributes. Beyond offline accuracy, deployment alongside a municipal cadastre imposes constraints rarely addressed in prior work: training and inference must share the

cadastral schema, the official territorial vocabulary, and any locational signal the model relies on.

Prior Brazilian work establishes ensemble methods on listing data but focuses on offline accuracy [Afonso et al. 2019, Marzagão et al. 2021, Sousa Neto and Ladeira 2022, Uriona-Maldonado et al. 2024]. This leaves open the requirements of a system operating alongside a municipal IPTU cadastre: parity with the cadastral schema, standardization against official shapefiles, and robust handling of high-cardinality territorial categories.

To bridge this gap, we present a production-oriented machine learning pipeline for mass property valuation, developed in partnership with the municipal government of Goiânia, Brazil (Prefeitura de Goiânia). The pipeline integrates three contributions. First, a cadastre-compatible feature design: training features are restricted to attributes available or derivable from the IPTU cadastre at inference time, while listing-only fields support cleaning but never propagate to the model, by construction preventing a class of training-serving skew underexplored in prior Brazilian work. Second, spatial standardization through official sources: coordinates are joined against the municipal shapefile, yielding standardized neighborhood, micro-region, and macro-region identifiers with a fallback category outside shapefile coverage, so the territorial vocabulary at training matches the one exposed in production. Third, a semantic neighborhood grouping from a large language model (LLM): to address the long tail of more than 500 standardized neighborhoods, Gemini 2.5 Pro is used once, offline, to classify each into one of ten locational market tiers; the mapping is persisted as a versioned artifact and applied deterministically at inference, with the LLM never invoked in the serving path.

On approximately $35{,}000$ deduplicated listings from Goiânia, CatBoost is selected from three tree-based ensembles and reaches a holdout mean absolute error (MAE) near BRL $222{,}000$ to $225{,}000$, R^2 around 0.89, and a median absolute percentage error (MdAPE) close to 11%. Across ten locational feature subsets, four pre-committed candidates land within 1.5% of each other on holdout MAE, a small family of competitive designs along a parsimony-robustness axis rather than a single optimum; coordinates combined with the LLM-derived group match the full baseline using fewer features.

2. Related Work

The proposed work draws on three lines: machine learning for mass appraisal, spatial feature engineering, and language models as feature engineering tools.

Property valuation is commonly grounded in hedonic price theory, where prices are a function of structural, locational, and contextual attributes [Rosen 1974]. Traditional mass appraisal relies on linear or semi-linear hedonic models, but real estate data exhibit nonlinear interactions among area, property type, location, and construction patterns, which tree-based models capture without an explicit functional form. Brazilian evidence supports this choice. Oliveira *et al.* [Oliveira et al. 2022] benchmarked tree-based methods for land valuation in Fortaleza; Marzagão, Ferreira, and Sales [Marzagão et al. 2021] reported competitive Random Forest results on web-scraped listings from central-west Brazil; Afonso *et al.* [Afonso et al. 2019] modeled more than 12 million Brazilian advertisements with Random Forest and deep learning; and Uriona-Maldonado, Vaz, and Zaghi [Uriona-Maldonado et al. 2024] obtained their lowest errors with XGBoost over features similar to those adopted here. International

studies report comparable results with tree-based mass appraisal and geocoded spatial features [Iban 2022, Tchuente and Nyawa 2022], which motivates a focused benchmark on Random Forest [Breiman 2001], XGBoost [Chen and Guestrin 2016], and CatBoost [Prokhorenkova et al. 2018].

Location is one of the main determinants of property value, but its representation in automated valuation models (AVMs) varies substantially. Raw latitude and longitude encode point position but not administrative boundaries, planning zones, or the spatial units used by municipal staff. Spatial information improves price estimation: Sousa Neto and Ladeira [Sousa Neto and Ladeira 2022] used spatial components from location APIs in a Brazilian public-sector valuation setting, and Tchuente and Nyawa [Tchuente and Nyawa 2022] showed gains from geocoded features. These studies leave open the operational mismatch between listings and municipal cadastres: listing neighborhoods are often misspelled, informal, outdated, or replaced by development names, whereas cadastres use official territorial units, so a model trained on listing strings can learn spatial patterns that cannot be reproduced at inference. The present work instead treats spatial feature engineering as geographic standardization against official shapefiles.

Recent work explores textual and language-based representations in real estate valuation. At BRACIS 2022, Bittencourt *et al.* [Bittencourt et al. 2022] showed that listing text can add signal beyond structured attributes, and Baur, Rosenfelder, and Lutz [Baur et al. 2023] reached a similar conclusion using property descriptions. More recent work uses LLMs to extract structured attributes from advertisements, such as construction standard and renovation indicators [Faxvaag et al. 2026], while attempts to use LLMs as direct appraisers remain less competitive than specialized ML models [Geerts et al. 2025]. The present work follows the feature-engineering interpretation: the LLM is not a price estimator but encodes standardized neighborhoods into locational tiers, turning urban semantic knowledge into a structured feature for long-tailed and newly appearing neighborhood categories.

3. Proposed Method

The proposed pipeline ingests property records compatible with the municipal cadastral schema and estimates market value through tree-based regression, formulated as supervised regression. Given property records $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, where $\mathbf{x}_i$ denotes the cadastre-derivable attributes of property i and $y_i \in \mathbb{R}^+$ its advertised market price, the goal is to learn $f_\theta \colon \mathbf{x} \mapsto \hat{y}$ from attributes available at inference time by minimizing MAE,

$$\theta^\star = \arg\min_\theta \frac{1}{n} \sum_{i=1}^{n} \big| y_i - f_\theta(\mathbf{x}_i) \big|. \tag{1}$$

The advertised price observed at scale on listing platforms is the market signal available for training, and modeling it directly provides a current, market-driven basis for mass appraisal. Figure 1 summarizes the end-to-end design: training and inference share standardization artifacts, and the language model is invoked only offline. The five stages are detailed below.

3.1. Dataset and Production Constraints

Training data were collected from property sale listings on VivaReal, a Brazilian real estate portal, for the municipality of Goiânia during September 2025. The crawler retained all

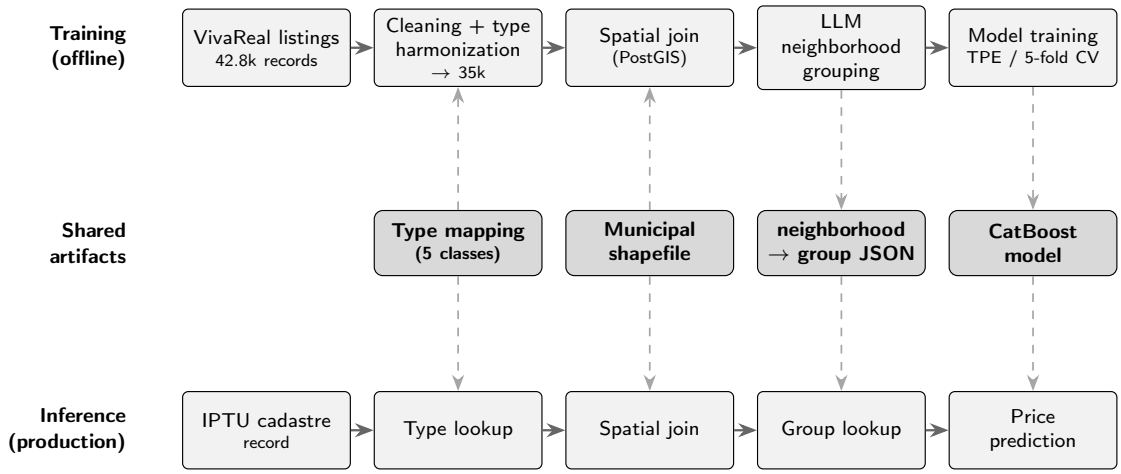


Figure 1. End-to-end pipeline for cadastre-compatible mass property valuation. The training lane (top) turns VivaReal listings into a CatBoost model; the inference lane (bottom) scores IPTU cadastral records. Both lanes share four artifacts: the five-class type mapping, the municipal shapefile, the neighborhood-to-group JSON, and the trained model. The LLM is not invoked in the serving path.

property categories, and a first deduplication step kept only the most recent listing per `id_imovel`, the platform-level identifier. Missing coordinates were enriched via the Google Maps Geocoding API, yielding $42{,}846$ records as the starting point of the pipeline. The collection therefore represents a contemporaneous market snapshot; operational use is expected to refresh the fitted model as new listing cohorts accumulate.

A central design constraint is the asymmetry between training and inference feature sets [Sculley et al. 2015]. At inference, the system receives records from the municipal IPTU cadastre, which exposes a smaller schema: property type, total area, usable area, and coordinates derived from the cadastral registration. We therefore distinguish two roles for the fields available in listings. Modeling features have a counterpart in the cadastral schema or a stable derivation path from it (property type, usable area, total area, latitude, longitude, and the territorial identifiers derived from the spatial join in Section 3.3) and compose the input vector $\mathbf{x}$. Auxiliary fields (bedrooms, bathrooms, suites, parking, listing title, free-text description, and the listing-reported neighborhood string) are richer in listings than in the cadastre and used only for data quality assessment, deduplication, and label correction; they are not propagated to the model. Remaining listing fields are discarded as too sparsely populated, lacking a stable cadastral counterpart, or out of scope for a structured tabular model (*e.g.*, `taxa_condominio`, `rua`, raw images).

3.2. Data Cleaning and Property Type Harmonization

Web-scraped listings exhibit heterogeneity, missingness, textual inconsistency, and redundancy, so cleaning proceeds as a sequence of conservative stages that prioritize data quality over volume. We first restrict the scope to the cadastral use case by removing under-construction records (`status_imovel`), allotment land variants, and farms, since only IPTU-bearing built properties are within scope. Total area is imputed with usable area when missing, and inconsistent records are corrected so that total area is not smaller than usable area. Listings whose title or description mentions “ágio” (a partial-equity sale mode

common in Brazilian condominiums) and whose price is below BRL $200{,}000$ are removed, since the reported value in such cases typically refers to the equity transfer amount rather than the full property value.

Listing platforms expose subtypes finer than the cadastral schema, so VivaReal types and IPTU codes are collapsed into five classes: home, apartment, condominium, commercial, and storage/warehouse. Casa and Sobrado map to home, Condomínio to condominium, and VivaReal flat and penthouse to apartment. The freely filled `tipo_imovel` field still contains residual noise, so we apply two conservative corrections: a 26-entry gazetteer of high-end horizontal gated communities reassigns 671 listings (1.91%) to condominium when the neighborhood field carries the development name, and a title-only regex classifier corrects labels when class-specific lexicons indicate a mismatch. Descriptions are excluded because they often mention neighboring typologies and produce systematic false positives.

Finally, before any train/validation split, deduplication addresses the frequent republication of the same physical property. The pipeline first keeps the most recent record per `id_imovel`, ordered by insertion date; because the same property may reappear under a new `id_anuncio`, it then defines a composite fingerprint over (normalized description, bathrooms, bedrooms, suites, usable area, neighborhood, price), conditioned on a non-empty description, and again retains the most recent record. This procedure removes $3{,}212$ records (about 7.5%), reducing duplicated-property leakage across folds before model selection [Sculley et al. 2015]. After cleaning, the dataset contains approximately 35k listings. Table 1 reports the distribution of price by collapsed property type, showing that `condominium` and `shed_deposit_warehouse` concentrate higher prices and stronger dispersion than `home` and `apartment`, which motivates the stratification by property type adopted during cross-validation (Section 4).

Table 1. Price distribution by collapsed property type after cleaning. Counts (n) and prices in BRL.

Property type	n	Mean	Median	Std
`apartment`	19,185	1,064,268	710,400	1,055,374
`home`	8,461	843,104	700,000	625,474
`condominium`	5,002	3,296,565	2,900,000	2,279,806
`commercial`	2,004	1,636,678	900,400	1,967,743
`shed_deposit_warehouse`	393	2,180,433	1,380,000	2,426,144

3.3. Spatial Standardization

Coordinates are validated conservatively against permissive city limits, with latitude in $[-16.90, -16.50]$ and longitude in $[-49.55, -49.10]$. The same check identifies swapped latitude/longitude pairs, correcting 47 records; only 16 records still outside the ranges after correction are removed, so outright errors are discarded without rejecting borderline records at the city limits.

The spatial join projects the listing dataset and the cadastral inference dataset into a shared territorial vocabulary: in listings the neighborhood field is filled freely and exhibits spelling variants, abbreviations, and frequent confusion with development

names, whereas the cadastre uses official administrative names. By joining each property's coordinates against the official municipal shapefile (SIRGAS 2000, provided by the city administration) using PostGIS, the spatial extension of the PostgreSQL database[1], we replace the free-text neighborhood with the shapefile-supplied integer identifier and obtain a representation identical at training and inference, regardless of the textual quality of the source. The join also exposes two coarser hierarchical levels, `microregiao_id` and `macroregiao_id`, which are not present in the listing data and provide more stable locational signals than the high-cardinality `bairro_id` (> 500 categories in the shapefile).

The shapefile is incomplete near city borders. Rather than discard records that may occur at inference, unresolved territorial identifiers are assigned identifier 0, reserved outside the official code space. Rare-category grouping for the long-tailed `bairro_id` and `microregiao_id` distributions is evaluated empirically (Section 4).

3.4. LLM-Assisted Neighborhood Grouping

Locational signal is among the strongest predictors of property value, but the standardized neighborhood identifier has practical limitations: after spatial standardization the city still has more than 500 neighborhoods with a long-tailed frequency distribution, which fragments the categorical support and weakens generalization on rare cells. We therefore introduce a semantic grouping of neighborhoods supported by a large language model.

We use Gemini 2.5 Pro as a feature engineering aid. The model was selected for its coverage of Brazilian Portuguese and local urban knowledge, with temperature set to 0 for reproducibility and all other parameters at default. Because the step is offline, persisted as a JSON artifact, and manually audited, the protocol is robust to substitution by other frontier LLMs. The model is prompted to group standardized neighborhoods into ten ordinal locational profiles based on centrality, infrastructure, occupation pattern, valuation history, proximity to similar areas, and relative market position, plus a fallback group (-1) for unresolved cases. Lower numbers denote higher-valued areas, from elite and high-standard locations (groups 1–2) through middle-class areas at different consolidation stages (3–5) to lower-middle, peripheral, transitional, rural-condominium, and popular subdivision patterns (6–10). The complete neighborhood-to-group mapping is persisted as a versioned JSON artifact[2] and reused deterministically at inference, so the LLM is not invoked in the serving path. Output was inspected manually on a sample of neighborhoods to verify consistency with their actual market profile, with the LLM acting as a scalable substitute for manual urban classification rather than as a price estimator. The resulting tiers do not nest inside the official hierarchy: elite horizontal gated communities such as Aldeia do Vale, Alphaville Flamboyant, and the Jardins developments are assigned to group 1 by market profile rather than by administrative position, and group 9 collects chácaras and rural condominium belts, a profile with no counterpart at the neighborhood, micro-region, or macro-region levels. The LLM receives only standardized neighborhood names and the grouping criteria, with no access to listing labels, model residuals, or validation folds. The resulting artifact is therefore treated as an externally informed market-profile grouping, not as a causal ranking of neighborhoods.

[1] `https://postgis.net`

[2] Prompt template and final mapping: `https://github.com/Gustavoaluiz/prompt-neighborhood-grouper`.

The resulting feature, `grupo_bairro_num`, is used alongside the standardized `bairro_id` rather than replacing it: the former contributes a stable market-profile signal, the latter fine-grained spatial granularity, with their empirical contribution examined in Section 5.3.

3.5. Model Training and Optimization

The final input vector contains nine attributes: property type, total area, usable area, latitude, longitude, neighborhood ID, micro-region ID, macro-region ID, and the LLM-derived neighborhood group. Categorical features are passed to each model through its native categorical handling.

We benchmark three tree-based ensembles under identical preprocessing and validation: Random Forest [Breiman 2001], XGBoost [Chen and Guestrin 2016], and CatBoost [Prokhorenkova et al. 2018]. They cover bagging and gradient boosting, support mixed numerical and categorical tabular data, and meet the low-latency serving constraints of the target system. The optimization protocol is specified in Section 4.

4. Experimental Setup

Before any model selection or hyperparameter tuning, the cleaned dataset (Section 3.2) is split once into disjoint 90% training and 10% holdout partitions, stratified jointly by listing-price deciles and collapsed property type. The holdout supports no selection decision and is opened only once, for the final evaluation.

4.1. Validation, Metrics, and Optimization

All pre-holdout evaluations are performed exclusively inside the 90% training partition using the same stratified 5-fold cross-validation (CV). The CV strata combine price decile and property type, with combinations below 10 samples merged into a residual stratum; the same fold indices are reused across model selection, rare-category thresholds, and spatial ablations. Each run reports global metrics from concatenated out-of-fold predictions over the training partition.

Performance is reported with MAE, the optimization objective; root mean squared error (RMSE), to monitor large residuals; mean and median absolute percentage error (MAPE and MdAPE), to summarize percentage appraisal error with different sensitivity to outliers; and the coefficient of determination (R^2), for comparability with prior work. For n evaluation records with observed price y_i, prediction $\hat{y}_i$, and mean $\bar{y}$, the metrics are

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \qquad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{2}$$

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{y_i}, \qquad \mathrm{MdAPE} = \operatorname*{median}_i \frac{100\,|y_i - \hat{y}_i|}{y_i}, \tag{3}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \tag{4}$$

Hyperparameters are tuned with the Tree-structured Parzen Estimator (TPE) algorithm in Hyperopt, minimizing mean cross-validation MAE with 80 evaluations for

XGBoost and CatBoost and 60 for Random Forest. Tree counts are fixed outside the search: Random Forest uses 500 trees, XGBoost uses early stopping with patience 50 on a 10% per-fold internal validation set and a 5,000-round cap, and the final CatBoost `iterations` value is the average best iteration across an inner 5-fold cross-validation (5,870 in our run). The search covers depth, learning rate, regularization, subsampling, and categorical-handling parameters when available, including the CatBoost one-hot and CTR settings.

4.2. Selection and Holdout Rules

The campaign proceeds through model family selection, rare-category grouping for `bairro_id` and `microregiao_id`, spatial feature ablation, and final holdout evaluation. Each step depends on the previous one, and every decision before the final evaluation uses only cross-validation evidence from the training partition.

Model family selection considers multiple metrics: among configurations with competitive cross-validation MAE, we prefer the highest CV-R^2 and the smallest train-to-CV generalization gap. Because the gap is compared only across models that already attain competitive cross-validation error, a smaller gap reads as a more reliable generalization estimate rather than as a tuning artifact; native handling of high-cardinality categoricals is the tie-breaker. For rare-category and spatial ablation experiments, the optimized CatBoost configuration, fold indices, and non-spatial features are fixed, and only the studied variable changes.

To avoid holdout shopping, final candidates are pre-committed before opening the holdout. Spatial configurations are grouped by feature design philosophy, and the best representative of each group, together with the comprehensive baseline, is evaluated once on the holdout. Any disagreement between cross-validation and holdout evidence is treated as discussion material, not as a trigger for further selection.

5. Results and Discussion

This section reports the four experiments defined in Section 4.2. Each step conditions on the previous decisions, and only the final evaluation (Section 5.4) uses the holdout partition.

5.1. Model Family Selection

We compare the three tree-based families of Section 3.5 after independent hyperparameter optimization on the full feature set, under the protocol of Section 4.1. Table 2 reports the cross-validation metrics with the train-to-CV gap in R^2.

Table 2. Model family selection on the full feature set after hyperparameter optimization. Metrics from concatenated out-of-fold predictions; the gap reports $R^2_{\text{train}} - R^2_{\text{CV}}$. MAE in BRL.

Model	CV MAE	CV R^2	CV MdAPE	Train R^2	Gap
Random Forest	250,323	0.851	10.82	0.953	**0.103**
XGBoost	**238,278**	0.843	**10.18**	0.993	0.150
CatBoost	247,071	**0.855**	11.74	0.979	0.124

The three families fall within a 5% CV-MAE range but expose different accuracy-generalization trade-offs. Random Forest has the smallest train-to-CV gap and the highest MAE; XGBoost minimizes MAE and MdAPE but has the largest gap; CatBoost attains the highest CV R^2 and the second-lowest MAE, within 4% of XGBoost, with a smaller gap. Following Section 4.2, we select CatBoost as the base estimator because it combines competitive error with higher CV-R^2 and native handling of high-cardinality categorical territorial features.

5.2. Rare-Category Grouping

After spatial standardization (Section 3.3), `bairro_id` contains 545 categories and `microregiao_id` contains 81. Under the fixed-estimator rule (Section 4.2), categories below a support threshold are mapped to a single residual code. We tested neighborhood thresholds of 15, 25, 35, and 50 observations with the optimized CatBoost model and common folds. Cross-validation MAE selected 25, reducing `bairro_id` to 167 effective categories; the threshold of 50 collapsed it to 99 categories and lost spatial information. With this representation fixed, we tested micro-region thresholds of 15 and 25, selecting 15 and reducing `microregiao_id` from 81 to 78 effective categories. The smaller effect for `microregiao_id` is consistent with its less fragmented support.

5.3. Spatial Feature Configuration

We isolate the spatial design through feature-subset comparisons. All runs use the selected CatBoost estimator, the rare-category thresholds from Section 5.2 (25 for `bairro_id`, 15 for `microregiao_id`), and the same folds; only the spatial subset varies, with property type, total area, and usable area fixed. Coords denotes latitude and longitude, and Hierarchy the micro- and macro-region identifiers. Table 3 reports ten configurations ordered roughly by predictive performance.

Table 3. Spatial feature configuration experiment. MAE in BRL.

Spatial feature subset	CV MAE	CV R^2	CV MdAPE
No spatial features	334,876	0.779	17.89
Neighborhood ID only	284,836	0.824	14.06
LLM group only	302,704	0.805	15.01
Administrative hierarchy (Neighborhood ID + Hierarchy)	268,907	0.842	13.09
All categorical (no coords)	267,519	0.843	13.01
Coordinates only	245,707	0.851	11.48
Coords + macro region	243,745	0.853	**11.26**
All except Neighborhood ID	243,033	**0.858**	11.53
Coords + LLM group	**241,751**	**0.858**	11.27
All spatial features	247,071	0.855	11.74

Spatial information drives most of the gain: removing it raises CV MAE by 35% relative to the best subset. Coordinates are the main component, since a coordinate-only model is already within 2% of the leader and removing coordinates from the full design costs 11% in CV MAE. In contrast, `bairro_id`, the LLM-derived group, and the administrative hierarchy are weaker without coordinates. Once coordinates are present, a single

aggregated locational feature is enough: Coords + LLM group, Coords + macro region, and All except Neighborhood ID match or improve on the full spatial baseline, suggesting limited marginal value from the high-cardinality neighborhood identifier. The rare-category experiment supports the same reading: grouping low-frequency neighborhoods improves over the original high-cardinality representation.

Four subsets advance to the holdout under Section 4.2 as the best representative of each design philosophy: All spatial features as the comprehensive baseline, All except Neighborhood ID as the pruned variant, Coords + macro region as the lean administrative option, and Coords + LLM group as the lean semantic option. Proximity to the leading CV MAE is not the selection criterion here: All spatial features sits 2.2% above the leader and is carried forward as the incumbent design, whereas Coordinates only, at 1.6%, is closer to the leader but was not pre-committed, since the pre-commitment covered designs that add an aggregated locational signal to coordinates. Its holdout behavior therefore remains untested, which we report rather than repair after the fact.

Coords + LLM group obtains the lowest CV MAE while using fewer features than the comprehensive baseline. Its closest non-LLM alternative, Coords + macro region, differs by only about 0.8% in cross-validation, so the ranking should be interpreted within the evaluated design space. The operational distinction is maintenance: administrative regions absorb updates through the spatial join, whereas the LLM-derived mapping routes unseen neighborhoods to fallback until the offline artifact is refreshed.

5.4. Final Evaluation on Holdout

The four subsets advanced from Section 5.3 are evaluated once on the holdout, each refit on the full training partition with the selected CatBoost estimator and rare-category thresholds, varying only the spatial features. Table 4 reports the metrics.

Table 4. Holdout performance of the four pre-committed feature design configurations. MAE and RMSE in BRL.

Spatial feature subset	MAE	R^2	MdAPE	MAPE	RMSE
All spatial features	224,685	**0.894**	10.99	21.30	**505,060**
All except Neighborhood ID	**221,887**	**0.894**	10.76	**20.93**	505,643
Coords + macro region	225,187	0.886	10.79	21.26	524,061
Coords + LLM group	223,791	0.889	**10.68**	**20.93**	516,650

The holdout results confirm the narrow cross-validation band: all candidates fall within 1.5% on MAE and 0.8 percentage points on R^2, so they should be read as competitive rather than as a strict ranking. All except Neighborhood ID obtains the lowest MAE, Coords + LLM group the lowest MdAPE, and the comprehensive baseline (All spatial features) the highest R^2. Because prices span heterogeneous segments (Table 1), the absolute MAE should be interpreted with percentage errors: a holdout MAE near BRL $222{,}000$–$225{,}000$ coexists with MdAPE near 11%, indicating that the median relative error is close to one tenth of the advertised price despite high price dispersion. Holdout errors are lower than the cross-validation estimates because each candidate is refit on the full training partition before evaluation, whereas the cross-validation metrics aggregate out-of-fold predictions trained on smaller folds. Since all spatial comparisons use a

fixed estimator and fixed folds, the ablation gives a deployment-relevant feature-subset contribution analysis rather than a globally optimal subset, which could favor different hyperparameters.

6. Conclusion

We presented a production-oriented mass-appraisal pipeline that aligns training and inference with a Brazilian municipal cadastre through cadastre-compatible features, official-shapefile spatial standardization, and an offline LLM-derived neighborhood group. Using the selected CatBoost model, four pre-committed spatial designs reached holdout errors within 1.5% on MAE and 0.8 percentage points on R^2 over roughly $35{,}000$ Goiânia listings. Coordinates carry most of the locational signal, and once they are present any aggregated locational encoding, whether the LLM group, the macro region, or the full hierarchy, performs equivalently within the observed spread, so the high-cardinality neighborhood identifier is dispensable. The contribution of the LLM group is therefore operational rather than accuracy-driven: it supplies that aggregated signal without depending on the cadastral hierarchy, at the lowest CV MAE and holdout MdAPE among the pre-committed designs.

Limitations. New neighborhoods are routed to a fallback group until the offline mapping is refreshed, and reliability may vary in areas with few comparable listings. Local use should therefore monitor cadastral data quality drift, shapefile coverage, sample support, and segment-level error diagnostics. The protocol transfers to municipalities with cadastral schemas and official shapefiles, but rare-category thresholds and neighborhood groups remain local; transfer across urban configurations remains open.

Future Work. Future work will assess SHAP-based attributions and LLM-generated explanations for contestability, extract cadastral physical attributes from listing images during training, and add IBGE census-tract indicators for neighborhood-level context.

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